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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**Machine Learning Approaches for E-Commerce Sales Forecasting**	
S. No.	**University ID**	**Name**
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Abstract

The rapid growth of e-commerce has generated large volumes of sales data, making accurate sales forecasting an essential task for businesses. Predicting future sales helps organizations optimize inventory management, reduce operational costs, improve customer satisfaction, and make informed business decisions. Traditional forecasting methods often struggle to capture complex patterns in sales data, whereas machine learning techniques can effectively analyze historical data and identify hidden trends.This project focuses on developing a machine learning-based sales forecasting system using historical e-commerce sales data. The project begins with data collection and preprocessing, including handling missing values, removing duplicate records, encoding categorical variables, and feature scaling. Exploratory Data Analysis (EDA) is then performed to understand customer purchasing behavior, seasonal trends, and the factors influencing sales.Several machine learning algorithms, including Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor, are implemented and compared to determine the most accurate forecasting model. The performance of these models is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. Visualizations such as sales trend graphs, heatmaps, correlation matrices, and feature importance charts are used to present the analysis and results.The expected outcome of this project is a reliable sales forecasting model that enables businesses to predict future demand accurately. The proposed system can support inventory planning, marketing strategies, pricing decisions, and overall business growth by providing data-driven insights for effective decision-making.

Signature of the Guide	HOD